Hess's Law and Enthalpies of Formation

Answer key with worked solutions

A. Understanding

1. Hess's law.

> The enthalpy change of a reaction is **the same whatever route** is taken from reactants to products. It is true because enthalpy is a **state function** (it depends only on the initial and final states), which follows from the conservation of energy.

2. $\Delta H_f^\circ = 0$.

> **$O_2(g)$, $N_2(g)$, Fe(s), C(graphite)**: elements in their standard states. $O_3(g)$ is not the standard state of oxygen; $H_2O(l)$ and $CO_2(g)$ are compounds.

B. Combining equations

3. Nitrogen dioxide.

> (a) The equations add directly; 2NO cancels: $\Delta H = +182.6\text{ kJ} + (-116.2\text{ kJ}) =$ **+66.4 kJ**. (b) This equation forms 2 mol of NO_2 from its elements, so $\Delta H_f^\circ(NO_2) = +66.4\text{ kJ} \div 2\text{ mol} =$ **+33.2 kJ/mol**.

C. Using enthalpies of formation

4. Three reactions.

> (a) $\Delta H^\circ = (1\text{ mol})(-84.0\text{ kJ/mol}) - [(1\text{ mol})(+52.4\text{ kJ/mol}) + 0\text{ kJ}] =$ **−136.4 kJ**
>
> (b) $\Delta H^\circ = [(1\text{ mol})(-634.9\text{ kJ/mol}) + (1\text{ mol})(-393.5\text{ kJ/mol})] - (1\text{ mol})(-1207.6\text{ kJ/mol}) = -1028.4\text{ kJ} + 1207.6\text{ kJ} =$ **+179.2 kJ** (endothermic: limestone must be heated strongly)
>
> (c) $\Delta H^\circ = [(1\text{ mol})(-1675.7\text{ kJ/mol}) + 0\text{ kJ}] - [(1\text{ mol})(-824.2\text{ kJ/mol}) + 0\text{ kJ}] =$ **−851.5 kJ** (very exothermic: it melts the iron)

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5. Propane.

> Products: (3 mol)(−393.5 kJ/mol) + (4 mol)(−285.8 kJ/mol) = −1180.5 kJ − 1143.2 kJ = −2323.7 kJ
>
> Reactants: (1 mol)(−103.8 kJ/mol) + (5 mol)(0 kJ/mol) = −103.8 kJ
>
> ΔH° = −2323.7 kJ − (−103.8 kJ) = **−2219.9 kJ**

6. Ammonia oxidation.

> Products: (4 mol)(+91.3 kJ/mol) + (6 mol)(−241.8 kJ/mol) = 365.2 kJ − 1450.8 kJ = −1085.6 kJ
>
> Reactants: (4 mol)(−45.9 kJ/mol) + (5 mol)(0 kJ/mol) = −183.6 kJ
>
> ΔH° = −1085.6 kJ − (−183.6 kJ) = **−902.0 kJ**

7. Vaporization of water.

> ΔH° = (1 mol)(−241.8 kJ/mol) − (1 mol)(−285.8 kJ/mol) = **+44.0 kJ**. This is the **enthalpy of vaporization** of water at 25 °C: the heat needed to turn 1 mol of liquid water into vapour.

D. Challenge

8. ΔH_f° of ethanol.

> $\Delta H^\circ = [2\Delta H_f^\circ(CO_2) + 3\Delta H_f^\circ(H_2O(l))] - [\Delta H_f^\circ(\text{ethanol}) + 0]$, so
>
> $\Delta H_f^\circ(\text{ethanol}) \times (1\text{ mol}) = (2\text{ mol})(-393.5\text{ kJ/mol}) + (3\text{ mol})(-285.8\text{ kJ/mol}) - (-1366.8\text{ kJ})$
>
> = −787.0 kJ − 857.4 kJ + 1366.8 kJ = −277.6 kJ, so ΔH_f°(ethanol) = **−277.6 kJ/mol** (matching the table)

9. Glucose.

> Products: (6 mol)(−393.5 kJ/mol) + (6 mol)(−285.8 kJ/mol) = −2361.0 kJ − 1714.8 kJ = −4075.8 kJ
>
> ΔH° = −4075.8 kJ − (1 mol)(−1273.3 kJ/mol) = **−2802.5 kJ** (per mole of glucose)
>
> Energy from 10.0 g: $\frac{10.0\text{ g}}{180.16\text{ g/mol}} = 0.05551\text{ mol}$; 0.05551 mol × 2802.5 kJ/mol = **156 kJ** released

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